MIL-C-22931/14
10 February 1968
SUPERSEDING
MIL-C-22931/3A (in part)
27 March 1964

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY,

SEMIRIGID, COAXIAL, SEMI-AIR-DIELECTRIC,

1.625 INCHES OUTSIDE DIAMETER, 75 OHMS

The complete requirements for procuring cables described herein shall consist of this document and the latest issue of Specification MIL-C-22931.

REQUIREMENTS:

Operating temperature range: -55 to +80° C.

Inner conductor: Copper wire or tube.
Overall diameter 0.374 (9.50 mm) ± 0.002 (.05 mm) inch.

Dielectrics:
Cable core and insulating layer: Polystyrene.

Outer conductor: Aluminum tubing, smooth, seamless.
Inside diameter: 1.472 (37.39 mm) inches.
Outside diameter: 1.625 (41.28 mm) inches.

Jacket: See table.

Voltage withstand:
6,000 volts rms (nominal)
11,000 volts dc.

Insulation defects, spark: See table.

Attenuation (decibels (dB) per 100 ft maximum):
0.13 dB at 30 MHz.
0.54 dB at 400 MHz.
0.95 dB at 1,000 MHz.

Velocity: 90.9 percent, nominal.

Capacitance: 14.7 pf/ft, nominal.

Impedance: 75 ± 1 ohms.

Voltage standing wave ratio:
Initial - 1.15 maximum for 10-1,000 MHz.
Following bending and temperature cycling tests - 1.20 maximum for 10-1,000 MHz.

Mandrel size for cold bend, bending, and temperature cycling tests: 40 inches (1016 mm).

Minimum bending radius - 25 inches (635 mm).

FSC 6145

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Requirement 1/	M22931/14 dash number	
	-001	-002
Jacket- - - - - - - - - - - - -	Not applicable	Polyethylene
Diameter - - - - - - - - - - -		1.735(44.07) +0.030(.76) -0.014(.36)
Minumum wall thickness - -		0.049(1.24)
Insulation defects, spark - -	Not applicable	Applicable

1/ Dimensions are in inches, millimeters are in parentheses. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4mm.

NOTES:

1. Cross reference information is as follows:

		Former classification	
Part number	RG number	Class	Type
M22931/14-001	RG248/U	N	I
M22931/14-002	RG249/U		

2. Figure is for information.

Custodians:
Army - EL
Navy - EC
Air Force - 11

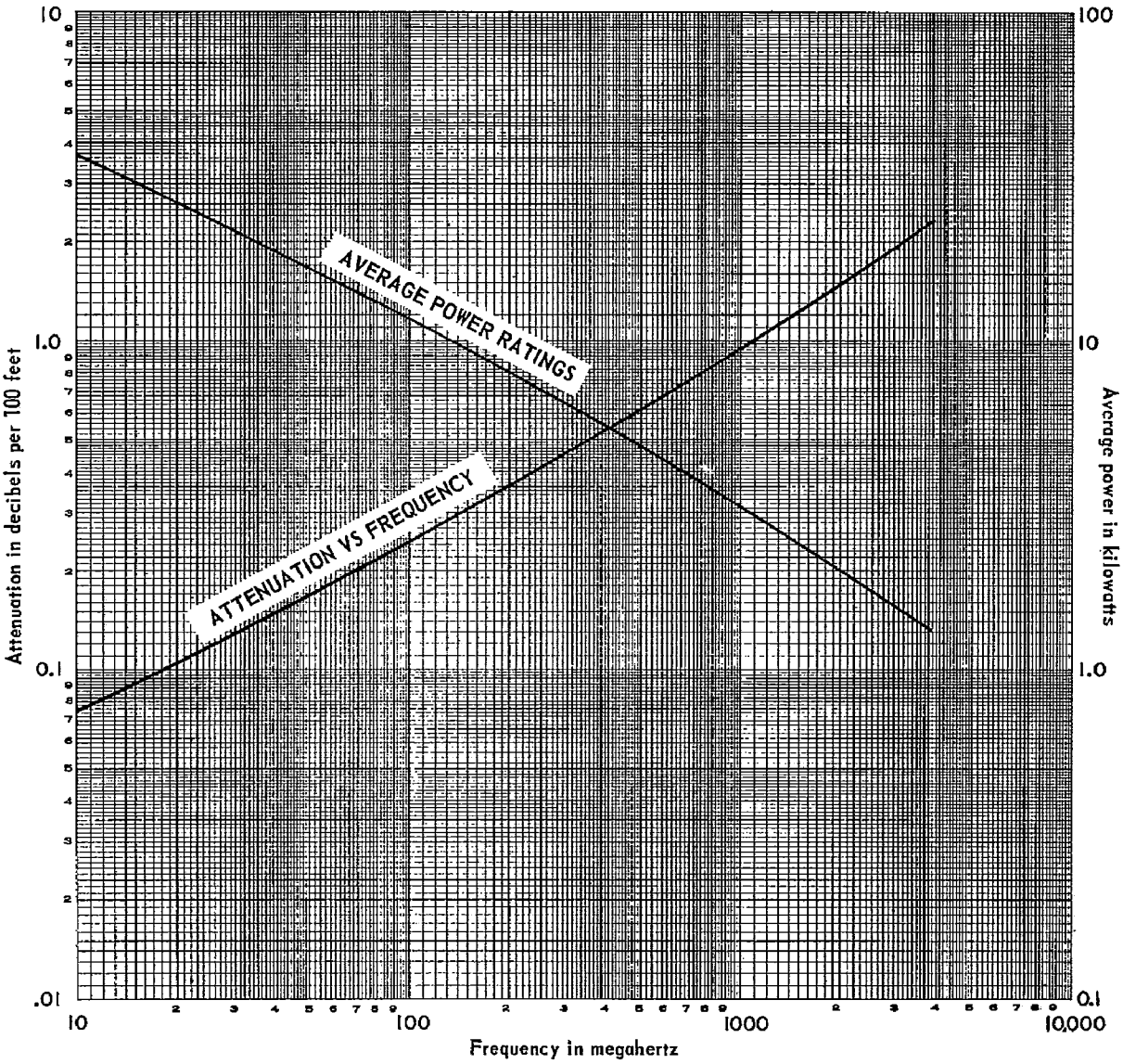
Review activities:
Army - EL, MU
Navy - SH, EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Navy - EC

(Project 6145-0431-3)

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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 1. Attenuation and power rating curves for -001 and -002.

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